HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **ABB AC500 PLC logic** using IEC 61131-3 standard languages for ship-to-shore crane automation, focusing on motion control sequences.
- Developed **Python scripts** to automate data extraction from production logs, reducing manual analysis time by approximately **40%** for operational reporting.
- Integrated **Java Spring Boot** REST APIs with existing enterprise systems, facilitating cross-service data exchange and enhancing system interoperability.
- Collaborated with cross-functional teams including electrical and mechanical engineers to troubleshoot and optimize control systems, ensuring safe and efficient crane operations.
- Wrote and maintained **production runbooks** and documentation for complex Java applications, decreasing incident resolution time for P1/P2 issues by **15%**.
- Configured and maintained **Jenkins CI/CD pipelines** for automated build, test, and deployment of Java applications, improving deployment frequency by **20%**.
- Containerized multiple microservices using **Docker**, enabling consistent deployment across development and production environments on **AWS EC2**.

SKILLS

PLC & Automation – ABB AC500, AC800M, IEC 61131-3, Automation Builder, Control Builder, Industrial Communication Protocols, Ethernet/IP, Modbus, Profibus, Profinet, HMI/SCADA, Motion Control, Safety Interlocks

Programming Languages – Java, Python, SQL, Shell Scripting

Frameworks & Libraries – Spring Boot, Hibernate, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile/Scrum, Linux, Splunk, Dynatrace

PROJECTS

Industrial Crane Automation Logic Refinement (HCL Technologies)

- Developed and tested **IEC 61131-3 PLC code** for a ship-to-shore crane control system, specifically enhancing safety interlock logic for container handling operations.
- Integrated PLC data with a prototype **HMI system** using Modbus TCP/IP for real-time status monitoring and diagnostic feedback, improving operator situational awareness.
- Analyzed performance logs from crane operations to identify bottlenecks, implementing code optimizations that reduced control loop execution time by **18%**.

Automated Production Reporting Pipeline (HCL Technologies)

- Engineered a **Python ETL pipeline** to extract, transform, and load operational data from PostgreSQL and Oracle databases into a central reporting schema.
- Automated the generation of weekly performance reports using **pandas** and scheduled their distribution via email, saving approximately **5 hours** of manual effort per week for the operations team.
- Implemented data validation checks within the pipeline to ensure data integrity, reducing downstream reporting errors by over **30%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)